

**Current Research Topics for USASMDC TC BAA For Science, Technology, and
Test and Evaluation Research
BAA Number: W9113M-24-R-BAA1
Issued: 26 Aug 2024**

**Current Research Topics for the
U.S. Army Space and Missile Defense Command
(USASMDC), Space and Missile Defense Technical Center
(SMDTC) Broad Agency Announcement (BAA) for Science,
Technology, and Test and Evaluation Research**

BAA Number: W9113M-24-R-BAA1

Disclaimer

All current SMDTC topics of interest can be found at:
<https://www.smdc.army.mil/ORGANIZATION/TC/>. Changes to these topics will be made
using the website on an as needed basis. This document is a printed copy of the current
SMDTC topics of interest as of the noted print date. Interested parties are encouraged
to continually browse the SMDTC webpage and review the SMDTC BAA for instructions
on submissions.

The SMDTC Broad Agency Announcement for Science, Technology, and Test and
Evaluation Research, W9113M-24-R-BAA1, is available on <https://www.grants.gov/> and
<https://sam.gov/>.

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Available SMDTC Topics of Interest

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USASMDC SPACE AND MISSILE DEFENSE COMMAND TECHNICAL CENTER
TOPIC OF INTEREST

Title: Development of Multiscale Electromagnetic Models for Susceptibility and Thermal Analysis

Announcement ID: TCBA001

SMDTC Office: SMDC-TCT-R

SMDTC Competencies: Test and Evaluation, Strategic Weapon Technologies, and Hypersonic Defeat

Key Words: HPM, modeling, electromagnetic

Description:

The USASMDC high-powered microwave (HPM) Team is interested in the development of Novel electromagnetic (EM) models to support our HPM susceptibility testing in both the macro and micro scale regions. It should have the ability to provide methods which promote self consistency in the transition between areas of scale such as the transition from submicron solid state to circuit level scale features. They should facilitate the calculation of emergent behaviors and aid in the discovery of new system behaviors.

Models should enable the calculation of effects at the system, device, and connection scales and allow for the modeling of dynamic thermal effects. Effects include things such as thermal stacking, energy storage considerations, and non-equilibrium thermodynamics should be capabilities of the modeling tools. The code should have the ability to aid in the design of experiments for model validation.

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USASMDC SPACE AND MISSILE DEFENSE COMMAND TECHNICAL CENTER
TOPIC OF INTEREST

Title: Development of New Electronic Warfare and RADAR Algorithms, Systems, and Models.

Announcement ID: TCBA002

SMDTC Office: SMDC-TCT-R

SMDTC Competencies: Test and Evaluation, Strategic Weapon Technologies, and Hypersonic Defeat

Key Words: None.

Description:

The USASMDC radio frequency (RF) Technologies Team is interested in the development of novel new Electronic Warfare and RADAR techniques utilizing multi-aperture transmitters and receivers to promote battlefield survivability and anonymity.

As the Army prepares for the battlefield of 2040 in which denial of asset location to the enemy becomes problematic, there exists a need for the development of new, inexpensive multi-aperture radar and electronic warfare systems where the transmission and reception of the transmitted signal is distributed over a wide geographical area. This work would include the development of new algorithms to enable better utilization of the capabilities, development of the hardware systems themselves, and creation of digital twins for simulation and evaluation purposes. Proposers do not have to address all 3 areas but should submit to their areas of expertise.

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TOPIC OF INTEREST

Title: Development of Novel Antenna Concepts

Announcement ID: TCBA003

SMDTC Office: SMDC-TCT-R

SMDTC Competencies: Test and Evaluation, Strategic Weapon Technologies, and Hypersonic Defeat

Key Words: HPM, antenna, wide band, time response

Description:

The USASMDC Team is interested in the development of novel high-powered microwave (HPM) antennas. The antennas should be capable of handling gigawatts of peak radio frequency (RF) input power and kilowatts of average RF input power. It should be capable of transmitting wide band signals and designed using time-based antenna modeling methods.

Typical minimum pulse widths to be supported should be 10ps or longer with an input voltage of 30kV. Typical feed line impedances should be 50 ohms with consideration to other impedances on a case-by-case basis. The desired features include high tolerance of repetition rates, frequency, and the use of Institute of Electrical and Electronics Engineers Standards Association (IEEE) standard or commercial off the shelf interfaces.

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TOPIC OF INTEREST

Title: Development of Novel RF Component Technologies

Announcement ID: TCBA004

SMDTC Office: SMDC-TCT-R

SMDTC Competencies: Test and Evaluation, Strategic Weapon Technologies, and Hypersonic Defeat

Key Words: HPM, connectors, microwave, high voltage

Description:

The USASMDC Team is interested in the development of novel high-powered microwave (HPM) components, cables, and connectors. The offeror should not be bound by traditional 50 Ohm architectures but instead look at component impedances that maximize performance. Of particular importance is increasing the working alternating current (AC) and direct current (DC) voltages of components and connectors to allow for the safe use of connectors above 20 kV.

As the Army prepares to field HPM capabilities, the need arises to be able to connectorized subsystems in a manner that the Soldier can perform field maintenance, without requiring a proprietary component, cable, or connector. What is required is a series connectors and components that can be Government-owned and published as IEEE or mil-spec standards for the production of commercial off the shelf interfaces. Additionally, cabling that can support the higher electrical stresses are needed (50-100 kV) as well as high voltage components.

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TOPIC OF INTEREST

Title: Development of Novel RF Transmitter Systems

Announcement ID: TCBA005

SMDTC Office: SMDC-TCT-R

SMDTC Competencies: Test and Evaluation, Strategic Weapon Technologies, and Hypersonic Defeat

Key Words: None.

Description:

The USASMDC radio frequency (RF) Technologies Team is interested in the development of high-powered microwave (HPM) and Standard RF Transmitter systems for outdoor testing. The offeror should be prepared to develop full-scale demonstration systems based upon a variety of solid-state technologies. It should also include provisions for adding discrete and integrated receiver modules.

As the Army prepares to field HPM capabilities, the need arises to build test systems for use in assessing the outdoor survivability and usefulness of RF technologies. These test systems should include integrated prime power and cooling, antenna pointer tracker systems such that range assessments and effects can be established into a variety of targets, sensors for establishing fine track and target verification. Additionally, if the need arises, they should possess the ability to receive the transmitted signal via either integrated sensors or using remote sensors to aid in target track and location. Frequency/waveform diversity and wide bandwidth are desired.

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TOPIC OF INTEREST

Title: Development of Novel Solid State HPM Sources

Announcement ID: TCBA006

SMDTC Office: SMDC-TCT-R

SMDTC Competencies: Test and Evaluation, Strategic Weapon Technologies, and Hypersonic Defeat

Key Words: HPM, solid state, high repetition rate

Description:

The USASMDC high-powered microwave (HPM) Team is interested in the development of novel solid state HPM sources based upon Non Linear Transmission Line (NLTL), Diamond, Gallium Nitride (GaN), Germanium Sulfide (GeS) or other solid-state technologies. Of particular interest to USASMDC are NLTL and diamond direct conversion technologies.

The desired source features include high repetition rates coupled frequency, pulse width and power tunability, and the use of IEEE standard interfaces. Minimum pulse repetition rates should be greater than 100 kHz with a minimum/maximum frequency of 3 GHz. High average power coupled with high peak power are desirable. Minimum source output power should be 1 megawatt with a clear path to combine multiple sources for powering a phased array antenna.

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TOPIC OF INTEREST

Title: Digital Transformation for Directed Energy Weapon Systems

Announcement ID: TCBA007

SMDTC Office: TCE-ES

SMDTC Competencies: Strategic Weapon Technologies

Key Words: digital transformation, directed energy

Description:

This program emphasizes applied research focused on defining and developing standard interfaces for directed energy weapons and utilizing digital tools for hardware-in-the-loop analysis of plug and play components. For the past 10 years there have been great strides made in directed energy technologies and applications of directed energy to defeat current and emerging threats. High energy laser (HEL) systems are capable of focusing an immense amount of power into a single, focused beam. Though these technologies exist and are nearing a transition point, there is still much work to be done with respect to the affordability of the technologies and closely related to that, the modularity of components.

HEL weapons, with their low cost-per-shot ratio, have been promoted as an affordable way to defeat certain known and emerging threats; however, these HEL weapon systems are often made up of expensive, proprietary components with non-standard interfaces. This makes interchangeability difficult and costly. There is an immediate need to develop cost-effective, standard interfaces to enable modular component upgrades to existing systems. Once fully defined, developed, and demonstrated in a laboratory setting, the technical data for these interfaces needs to be published in digital format. New components can then be developed to plug and play with the standard interfaces. Electronic versions of new components can be incorporated into the existing digital models of the interfaces and fully analyzed prior to prototype component builds. This digital transformation will also allow for enhanced hardware-in-the-loop testing of new and upgraded components. This will in turn reduce program cost and schedule risk, while enhancing the technical performance and speeding the transition of these technologies to the warfighter.

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Title: Energy Based Weapon Systems and the Application of the Tactical Microgrid Standard

Announcement ID: TCBA008

SMDTC Office: TCT-RC

SMDTC Competencies: Strategic Weapon Technologies, and Hypersonic Defeat

Key Words: High Power Microwave (HPM), Laser, Energy, Microgrid

Description:

The emergence of directed energy weapon systems (DEWS) presents a shift in modern warfare, offering unparalleled precision, scalability, and sustainability compared to conventional kinetic weapons. To successfully integrate DEWS, there must be robust power management solutions and seamless integration with existing and new infrastructure. In this context, Tactical Microgrid Standards (TMS) offer a framework for optimizing power distribution, enhancing resilience, and facilitating interoperability within military environments. The TMS has emerged as a key enabler for enhancing energy flexibility and optimizing power distribution in military settings. This topic aims to explore the integration of TMS principles into emerging DEWS platforms, focusing on power management and the development of standardized interfaces to streamline energy weapon deployment and maximize operational effectiveness.

The TMS is currently being researched and designed for various military platforms. USASMDC would invite white papers on TMS-based power systems for transportable directed energy weapons (HPM and Light Amplification by Stimulated Emission of Radiation [LASER]). The proposed systems should be able to handle large random loads with extreme variations of output power being able to handle the sudden turn on of the directed energy loads (maximum dP/dt).

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TOPIC OF INTEREST

Title: Short and Ultrashort Pulsed Laser Research for Directed Energy Applications

Announcement ID: TCBA009

SMDTC Office: TCT-MDD-DER

SMDTC Competencies: Hypersonic Defeat

Key Words: laser directed energy, ultrashort pulse lasers, laser ablation, spectroscopy

Description:

The Pulsed Laser Lab (PuLL) in the Directed Energy Research Division seeks to discover innovative concepts for the application of short and ultrashort pulsed lasers (pulse durations less than a microsecond) to counter future threats. Given the recent rapid increases in the capabilities of these classes of lasers, novel engagements are emerging in the realm of practicability that would lead to technological overmatch.

General areas of interest in this topic include:

- Novel pulsed laser sources reaching new levels of average power, wide range of wavelengths, and increased repetition rates including burst mode operation;
- Nonlinear atmospheric propagation with ultrashort pulse lasers;
- Short and ultrashort pulsed laser ablation and surface modification;
- Advanced beam control techniques for improved beam quality and propagation of pulsed lasers over a wide range of powers levels and wavelengths; and
- Next generation diagnostic concepts for ultrashort pulse laser measurements.

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Title: Development of Novel Technologies for HPM and Hypersonic Systems and Testing.

Announcement ID: TCBA010

SMDTC Office: SMDC-TCT-R

SMDTC Competencies: Test and Evaluation, Strategic Weapon Technologies, and Hypersonic Defeat

Key Words: HPM, RF, hypersonic, radar, AI/ML, quantum sensors, light gas gun, hypervelocity

Description:

The USASMDC Weapons and Protective Technologies Division (WPTD) Team is interested in the development of high-powered microwave (HPM), radio frequency (RF), hypersonic, quantum sensor, and hypervelocity research, technology, and components. Technology development can be applicable to new or improved components, systems, and test facilities, and can include processes, and techniques. All aspects of the technology life cycle can be considered including design, development, manufacturing, testing and evaluation, insertion, sustainment, maintenance, and supply chain management.

Of particular interest to USASMDC are the following:

- HPM capabilities such lethality, destructive, and non-destructive (e.g. electromagnetic warfare);
- Hypersonic capabilities such as individual components and non-kinetic defeat capabilities;
- Quantum sensor technology;
- Light gas gun research such as hypervelocity research on impact and ballistics;
- Advanced manufacturing for components related to radar and HPM sources and components;
- Artificial intelligence and machine learning (AI/ML) algorithms, tools, and M&S related to all areas of research listed above and other WPTD relevant technologies; and
- Technology specific to WPTD test facilities including the HPM Effects Laboratory (HPMEL), the Aero-physics Research Facility (ARF), and the Hot Air Tunnel (HAT).